

Two hubs can clear the hurdle— if the operating assumptions hold

European service-hub expansion • Decision analysis of six markets • 2025 shipped sales, returns through 31 January 2026

RECOMMENDATION

Conditionally prepare **Czechia + Spain** for launch. Reserve €225,000 of year-zero capex and 5 FTE; release spend through the 90-day validation gates below.

BASE ANNUAL INCREMENT	JOINT STRESS	SIMPLE PAYBACK	2025 CONTRIBUTION
€198,099	€32,180	1.14 yrs	€2,030,010

Why this choice

At the client's 25% base volume uplift, the selected pair produces €198,099 a year after recurring fixed cost. It ranks first among 17 feasible single/pair choices, stays positive in the defined combined return-rate/FX stress (€32,180), and uses €225k / 5 FTE versus ceilings of €450k / 7. The strongest base-case alternative, Poland + Spain, gives €186,329 at €240k / 6 FTE. The model is assumption-led and the client transactions are synthetic; this is a conditional investment case, not causal proof. [S1–S2]

Board action requested

Approve conditional design and validation authority for CZE + ESP, with no irreversible lease or fit-out commitment until the day-30 gate confirms a ≥25% volume case and the stated savings per unit from route and handling evidence. Stop or defer if either market misses the decision threshold. The defer option remains the zero-contribution, zero-capex reference.

2025 base: 1,254 distinct eligible shipped orders • 77,436 units • €4,383,792 net sales • €2,030,010 contribution • 46.3% contribution / net sales. [S1, S2]

The synthetic baseline favors Spain, Czechia and Poland

The 1,254 eligible orders are balanced at 209 per market in this generated extract; higher unit volume and product mix drive most country differences. Spain and Czechia lead on contribution and contribution margin. That is evidence about this synthetic 2025 ledger, not evidence of real customer demand or local-hub causality. [S1]

Market	Orders	Units	Gross sales	Refunds	Net sales	Contribution	Margin
DEU	209	12,153	€631,245	€5,453	€625,792	€262,366	41.9%
FRA	209	11,880	€640,930	€8,699	€632,231	€276,271	43.7%
NLD	209	12,576	€709,109	€11,376	€697,733	€315,644	45.2%
POL	209	13,163	€776,508	€14,618	€761,890	€356,494	46.8%
CZE	209	13,733	€834,572	€19,265	€815,307	€397,335	48.7%
ESP	209	13,931	€872,259	€21,420	€850,839	€421,900	49.6%
Total	1,254	77,436	€4,464,622	€80,831	€4,383,792	€2,030,010	46.3%

Country share is not the same as opportunity. Hub savings, fixed cost, capex and headcount shift the ranking: the top baseline contributor is Spain; the best two-market arithmetic is Czechia + Spain. Refunds are highest in euros and units in Spain, but the decision model does not assume every returned unit recovers product cost; only recorded restocks do. [S1]

Macro context: broad, non-demand indicators

Market	Population 2024	Change 2022–24	GDPpc 2024 current US\$	GDPpc change 2022–24
DEU	83,516,593	0.4%	\$56,104	11.1%
FRA	68,551,653	0.5%	\$46,103	12.5%
NLD	17,993,485	1.7%	\$67,465	14.1%
POL	36,559,233	-0.7%	\$25,104	32.9%
CZE	10,905,028	2.2%	\$31,828	12.5%
ESP	48,848,840	2.2%	\$35,327	16.5%

Population growth is strongest in Czechia (+2.2%) and Spain (+2.2%); Poland declined about 0.7% over the period. 2024 GDP per capita ranges from roughly \$25.1k (Poland) to \$67.5k (Netherlands), while two-year nominal USD changes also reflect prices and exchange rates. Neither population nor current-US\$ GDP per capita directly measures replacement-assembly demand. Values preserve the archived World Bank response vintage (last updated 13 July 2026); see [S3–S4].

Ledger controls keep revisions, refunds and FX on the right grain

The analysis pools both order extract pages and the correction file, removes identical repeats, picks the highest numeric revision by order ID, then applies shipment eligibility. It selects the highest return revision by return ID, applies the 31 Jan cutoff and eligible-order link, and aggregates distinct return IDs additively. No conflicting winning revisions were detected. [S1]

Check	Observed	Treatment
Orders: 1,328 raw rows	13 identical rows removed; 1,297 IDs; 18 lower revisions superseded	1,254 eligible latest rows after 24 cancellations, 18 tests and one 2026 shipment excluded
Returns: 264 raw rows	20 identical rows removed; 243 IDs; 1 revision superseded	241 records applied; one 1 Feb return excluded; one orphan quarantined
Blank shipment dates	24 selected rows; all are cancelled	Recorded as missing but cannot qualify; no blank date on included shipment
Free/zero-gross shipments	12 valid orders	Included in orders and units; zero sales are not dropped
Currency and quantities	No country-currency mismatches, negative gross, refunds above gross or returned qty above shipped	All 24 PLN/CZK monthly means have 20–23 published business-day quotes
Country/month reconciliation	72 rows roll to 6 country totals	All nine required measures reconcile exactly; order-level contribution identity passes

EUR conversion follows the original shipment month. ECB quotes are local currency per EUR; local proceeds/refunds are divided by the monthly mean, not multiplied or inverted. The same sale-month mean translates later refunds. EUR transactions use 1. Each order's gross sales, summed refunds, net COGS and fulfillment are rounded to cents with ROUND_HALF_UP before aggregation. The 2025 daily ECB archive is an analytical translation proxy, not actual settlement FX. [S1–S2]

Booked revenue, cash and contribution

The workbook calls gross sales less credited refunds 'net sales' (€4,383,792); it is a transaction measure in this extract, not cash collected. Cash timing, actual settlement rates, taxes, chargebacks, payment fees and receivable/payable timing are absent. Contribution subtracts €2,246,919 net product cost and €106,862 nonrefundable fulfillment, yielding €2,030,010. COGS recovery is limited to recorded restocked units at the sale-date unit cost; no blanket recovery is assumed. [S1]

The return set contains 241 included distinct records; 1,081 physical units returned (not a return-order count). The orphan ID ABSENT / return R-ORPHAN is kept in the exception log with no guessed join. R-LATE, received 1 Feb 2026, is outside cutoff. Source rows and exceptions are retained under evidence/processed for audit.

Single hubs: baseline strength does not guarantee robust economics

The client policy applies low/base/high uplifts of 10% / 25% / 40% to 2025 baseline contribution C and shipped units U: $\text{annual increment} = C \times u + U \times (1+u) \times s - F$. Capex K is shown separately as year-zero cash; it is not deducted from annual contribution. Simple payback is $K / \text{positive annual increment}$. Stress is a 25% volume case plus a 3%-of-gross additional refund shock with no added COGS recovery and a 10%-of-net-sales depreciation shock only for PLN/CZK. [S1]

Single-market choices

Hub	Capex EUR	FTE	Fixed/y EUR	Low Δ/y EUR	Base Δ/y EUR	High Δ/y EUR	Stress Δ/y EUR	Base pb (years)
DEU	€280,000	5	€95,000	-€34,006	€10,089	€54,183	-€13,583	27.75
FRA	€210,000	4	€85,000	-€19,476	€27,133	€73,741	€3,098	7.74
NLD	€140,000	3	€65,000	€10,832	€64,215	€117,598	€37,623	2.18
POL	€110,000	3	€45,000	€25,400	€83,612	€141,825	-€40,743	1.32
CZE	€95,000	2	€40,000	€31,457	€95,383	€159,309	-€37,827	1.00
ESP	€130,000	3	€55,000	€33,162	€102,716	€172,270	€70,007	1.27

Spain is the strongest single on base contribution (€102,716/year, €130k capex, three FTE); Czechia provides €95,383/year on €95k capex and two FTE. Germany's €10,089 base increment carries €280k capex and 27.75-year simple payback. Defer remains a €0 / 0 FTE reference.

CZE + ESP leads the additive pair cases

The 11 feasible pairs below satisfy both €450k capex and seven-FTE limits. Pair savings and costs are summed with no synergy. Four other pairs are excluded: DEU + FRA (€490k / 9 FTE), DEU + NLD (8 FTE), DEU + POL (8 FTE), and DEU + ESP (8 FTE).

Pair	Capex EUR	FTE	Fixed/y EUR	Low Δ /y EUR	Base Δ /y EUR	High Δ /y EUR	Stress Δ /y EUR	Base pb (years)
CZE + ESP	€225,000	5	€95,000	€64,619	€198,099	€331,580	€32,180	1.14
POL + ESP	€240,000	6	€100,000	€58,562	€186,329	€314,096	€29,264	1.29
POL + CZE	€205,000	5	€85,000	€56,856	€178,995	€301,134	-€78,570	1.15
NLD + ESP	€270,000	6	€120,000	€43,994	€166,931	€289,869	€107,630	1.62
NLD + CZE	€235,000	5	€105,000	€42,289	€159,598	€276,907	-€203	1.47
NLD + POL	€250,000	6	€110,000	€36,232	€147,828	€259,423	-€3,119	1.69
FRA + ESP	€340,000	7	€140,000	€13,687	€129,849	€246,012	€73,105	2.62
FRA + CZE	€305,000	6	€125,000	€11,981	€122,516	€233,050	-€34,729	2.49
FRA + POL	€320,000	7	€130,000	€5,924	€110,745	€215,566	-€37,645	2.89
DEU + CZE	€375,000	7	€135,000	-€2,549	€105,472	€213,492	-€51,410	3.56
FRA + NLD	€350,000	7	€150,000	-€8,644	€91,348	€191,339	€40,721	3.83

CZE + ESP is €11,770/year above the next-best base alternative (POL + ESP), has €2,916/year more stress contribution, and uses €15k less capex and one fewer FTE. Exact option-level assumptions and four paybacks per choice are in the workbook and metrics JSON. [S1]

CZE + ESP wins the stated policy case, but validation is the investment test

Selected case: Czechia + Spain uses €225,000 year-zero capex and 5 FTE, below the caps by €225,000 and 2 FTE. It yields €64,619 / €198,099 / €331,580 annual incremental contribution in low/base/high, and €32,180 in the defined joint stress. The base simple payback is 1.14 years; no discounting or ramp-up is included.

Why it beats the nearest alternative

Poland + Spain is the strongest base-case alternative at €186,329/year, €240,000 and 6 FTE, with €29,264 stress contribution and 1.29-year base payback. CZE + ESP is €11,770/year higher in base and €2,916/year higher in stress, while using €15k less capex and one fewer FTE. FX stress applies to both PLN and CZK sales: CZE + ESP includes a €81,531 CZK haircut on Czech net sales (Spain is euro-denominated) and remains at €32,180; POL + CZE absorbs €157,720 combined PLN/CZK haircut and falls to -€78,570. The recommendation therefore survives the stipulated CZK shock; it is not an FX-free case.

Decision pivot

For the selected pair, algebraic break-even uplift is about 2.7% under the given savings and annual fixed costs (vs 25% base). But this low break-even is not a measured demand estimate: all uplift is assumed. The board should change the decision to defer if route/handling trials cannot substantiate $\geq 25\%$ volume uplift and the given €2.10/unit CZE + €3.00/unit ESP savings, or if audited annual fixed cost rises enough to erase the stress cushion (€32,180). Test lower uptake, full ramp and no savings before final lease/capex release.

Cross-check against defer

Deferral retains €0 scenario contribution and avoids investment. The selected low case is +€64,619; the joint stress is +€32,180. The model therefore favors staged investment only if validation converts those assumed levers into operating evidence. No estimate is statistically proven or causal.

Risk register: uplift may not occur; central-hub savings may be counted twice or fail in local execution; rent, hiring and fit-out can exceed assumptions; added inventory can tie up cash; returned parts may not be reusable; FX reference stress is simplified; local tax, labor, customs, regulatory and service requirements are unmodeled. Owners and stop gates appear on page 6.

Use 90 days to turn synthetic assumptions into a gated operating case

Do not release lease or fit-out commitments at board approval. Start with reversible validation and location/design work, then approve each spend release against a named owner and evidence gate.

Days / gate	Accountable owner	Work and dependencies	Gate to proceed
0–30 Validate	COO (owner); CFO / Commercial Ops	Rebuild 2025 catchment baseline from transaction and route records; verify unit-level handling, parcel zones, transit, current central fulfillment cost, return/restock treatment, site/rent quotes and hiring plan. Dependency: access to actual non-synthetic operations data and candidate locations.	CFO signs audited baseline; ops demonstrates ≥25% feasible uplift case and savings ≥€2.10 CZE / €3.00 ESP per shipped unit; fixed-cost run-rate ≤€95k/y combined. Otherwise defer.
31–60 Design / pilot	Regional Ops (owner); Procurement / HR / IT	Pilot route consolidation or local stock positioning without permanent lease; collect real lead-time, on-time/in-full, cost/unit, stockout and return-restock data; secure conditional site and staffing quotes. Dependency: Gate 1, inventory master and service design.	COO + CFO approve site-level economics, service design, safety/quality and permits; modeled stress stays positive after quoted costs. Release only pilot/design budget.
61–90 Commit / launch readiness	COO (owner); CFO / HR / Quality / IT	Complete fit-out design, hiring, systems, inventory placement, supplier/3PL SLA, returns process, contingency and local compliance review. Dependency: Gate 2 approval and signed service-provider terms.	Board sponsor releases capex only when purchase orders and staffing fit €225k / 5 FTE envelope and KPI instrumentation is live. Otherwise pause or defer.

KPI plan: establish a 4-week matched central-ship baseline in each proposed catchment before any pilot; report weekly and monthly by country/SKU/order. Avoid treating total country growth as causal: use pilot vs matched routes/items where data permit and state limitations.

Measure	Definition / source	Gate / operating target	Owner / cadence
Volume uplift	Eligible shipped units vs matched 2025 baseline, adjusted for trend/seasonality	Run-rate ≥25% base assumption before full spend	Commercial Ops / weekly
Cost saved per unit	Avoided current fulfillment/handling less local variable cost ÷ units	≥€2.10 CZE; ≥€3.00 ESP	COO / weekly; Finance sign-off
Contribution after fixed cost	Incremental net sales – net COGS – fulfillment – recurring hub cost	Positive low-case and positive defined stress; base ≥€198k annualized combined	CFO / monthly
Service	Median and 90th percentile order-to-delivery days; on-time-in-full rate	Establish baseline by day 30; improve delivery P90 by ≥1 day and OTIF by ≥5 pp before rollout	Regional Ops / weekly
Returns / recovery	Refund € ÷ gross €; returned units; restocked units / returned units	No deterioration in refund rate; itemize recovery, no blanket credit	Quality / monthly
Spend / staffing	Committed and forecast capex; filled and planned FTE	≤€225k selected capex and ≤5 FTE; no unapproved recurring overrun	CFO + HR / monthly

If catchment-level baseline, route cost and actual customer/order geography cannot be obtained, do not proceed beyond reversible design work. The synthetic extracts do not include lead time, customer coordinates, stockouts or site cost evidence, so these must be measured before the gates can pass.

Evidence is useful for a structured choice; it cannot eliminate execution risk

The source room is a frozen local package. Its sales, returns, option costs, unit costs, staffing and scenario assumptions are synthetic (documented generator seed 2026092736). No interviews, live client systems or non-archived customer evidence were accessed. The archive's original source register and file hashes are preserved with the downloaded files. [S1–S4]

Material limitations

- Synthetic order rows cannot validate demand, market share, seasonality, churn or actual customer mix; equal 209-order totals per country are an artifact of this generated extract.
- No customer or delivery geography, promised/actual lead time, stockout, central-hub capacity, actual carrier route costs, service-level loss, lease quote, taxes, labor/regulatory cost or working-capital schedule is present.
- WB population and GDP per capita are national context only. GDP per capita is current US dollars (not PPP or constant-price income) and population is not parts demand.
- ECB monthly means are official reference rates, not transactional settlement FX. The 10% PLN/CZK shock is a stipulated scenario simplification, not a volatility distribution.
- Scenarios use annual steady-state additive arithmetic: no construction/ramp time, discount rate, inflation, tax, residual value, shared staffing, synergy, cannibalization or probability weights.
- Return rates, restocking and cost recovery follow recorded synthetic quantities. Physical reusability, warranty handling and future return timing may differ operationally.

Source register and direct links

[S1] **Synthetic client files** — data-dictionary.md; orders-part1.csv; orders-part2.csv; order-corrections.csv; returns.csv; unit-costs.csv; hub-options.csv; scenario-policy.md. Frozen source room downloaded 27 Sep 2026; see deliverables/source_register.csv for URL, exact local collection time and SHA-256.

[S2] **ECB historical reference-rate archive** — <https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip> — daily reference observations, archived file; source-room register retrieval 27 Sep 2026 06:36:40 UTC. Used only 2025 business days.

[S3] **World Bank population response** — https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000 — archived API payload; unit persons; archive response lastupdated 13 Jul 2026.

[S4] **World Bank GDP per capita response** — https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000 — archived API payload; unit current US dollars/person; archive response lastupdated 13 Jul 2026.

Companion files: Meridian_Analysis.xlsx for filtered data and charts; meridian_board_presentation.pdf for the board discussion; metrics.json for exchange; analysis/metrics.json and evidence/processed for audit; scripts/analyze.py and scripts/build_deliverables.py for reproduction.